PEMS03

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	STAEFormer	A2TTA (OL-AN)	A2TTA (STAE)
MAE	3	11.07 ± 0.16	10.40 ± 0.10	11.47 ± 0.16	10.88 ± 0.19	25.74 ± 1.07	9.71 ± 0.71	10.22 ± 0.04	9.69 ± 0.06	9.96 ± 0.16	10.33 ± 0.29	10.18 ± 0.18	10.94 ± 0.16	31.39 ± 1.10	10.03 ± 0.07	10.24 ± 0.39	12.35 ± 0.77	10.05 ± 0.13	10.54 ± 0.08	9.04 ± 0.05	*9.02 ± 0.03*	**8.48 ± 0.04**
	6	11.31 ± 0.12	11.33 ± 0.08	11.86 ± 0.14	11.53 ± 0.15	26.08 ± 1.05	10.38 ± 0.75	10.95 ± 0.05	10.40 ± 0.05	10.66 ± 0.17	11.01 ± 0.30	10.82 ± 0.13	11.95 ± 0.36	30.81 ± 2.24	10.73 ± 0.07	10.82 ± 0.41	13.23 ± 0.92	10.73 ± 0.12	11.50 ± 0.05	*9.38 ± 0.04*	9.70 ± 0.04	**8.86 ± 0.04**
	12	12.55 ± 0.11	13.40 ± 0.10	13.13 ± 0.14	13.14 ± 0.21	27.06 ± 0.83	11.76 ± 0.85	12.43 ± 0.09	11.80 ± 0.08	12.09 ± 0.28	12.47 ± 0.33	12.27 ± 0.12	14.10 ± 0.62	31.67 ± 4.70	12.16 ± 0.07	12.26 ± 0.48	15.30 ± 1.33	12.21 ± 0.09	13.69 ± 0.04	*9.99 ± 0.05*	10.95 ± 0.06	**9.48 ± 0.06**
	Avg	11.54 ± 0.13	11.53 ± 0.09	12.05 ± 0.15	11.72 ± 0.16	26.21 ± 0.98	10.49 ± 0.76	11.07 ± 0.04	10.50 ± 0.06	10.78 ± 0.19	11.13 ± 0.30	10.96 ± 0.15	12.15 ± 0.35	31.28 ± 2.42	10.84 ± 0.07	10.98 ± 0.42	13.46 ± 0.97	10.86 ± 0.12	11.72 ± 0.05	*9.42 ± 0.04*	9.78 ± 0.04	**8.88 ± 0.04**
RMSE	3	18.43 ± 0.22	17.29 ± 0.14	22.18 ± 0.56	17.93 ± 0.30	32.74 ± 1.15	16.05 ± 1.18	17.09 ± 0.17	16.10 ± 0.11	16.40 ± 0.14	16.82 ± 0.37	16.75 ± 0.23	17.77 ± 0.29	47.63 ± 1.51	16.58 ± 0.09	17.70 ± 0.67	19.87 ± 0.99	16.57 ± 0.18	17.56 ± 0.06	*15.11 ± 0.07*	15.11 ± 0.05	**14.19 ± 0.06**
	6	19.01 ± 0.15	19.19 ± 0.11	22.92 ± 0.54	19.21 ± 0.28	33.83 ± 1.08	17.47 ± 1.28	18.61 ± 0.18	17.54 ± 0.13	17.86 ± 0.19	18.32 ± 0.39	18.15 ± 0.16	19.93 ± 0.65	47.39 ± 3.49	18.05 ± 0.09	18.77 ± 0.69	21.36 ± 1.11	17.95 ± 0.15	19.62 ± 0.04	*15.84 ± 0.07*	16.49 ± 0.08	**14.98 ± 0.07**
	12	21.51 ± 0.12	23.23 ± 0.14	25.10 ± 0.50	22.34 ± 0.44	36.48 ± 0.82	20.23 ± 1.46	21.51 ± 0.20	20.25 ± 0.18	20.69 ± 0.39	21.30 ± 0.48	21.07 ± 0.20	24.35 ± 1.17	50.60 ± 7.75	20.91 ± 0.10	21.39 ± 0.80	24.93 ± 1.59	20.78 ± 0.09	23.96 ± 0.02	*17.04 ± 0.05*	18.87 ± 0.12	**16.19 ± 0.07**
	Avg	19.44 ± 0.16	19.53 ± 0.12	23.21 ± 0.54	19.56 ± 0.30	34.12 ± 0.99	17.66 ± 1.29	18.80 ± 0.17	17.71 ± 0.13	18.06 ± 0.22	18.53 ± 0.40	18.38 ± 0.19	20.32 ± 0.65	48.36 ± 3.82	18.24 ± 0.09	19.05 ± 0.71	21.76 ± 1.18	18.17 ± 0.14	19.99 ± 0.04	*15.88 ± 0.06*	16.59 ± 0.08	**14.99 ± 0.06**
MAPE (%)	3	33.31 ± 1.31	27.32 ± 0.50	31.78 ± 1.36	30.35 ± 0.75	57.38 ± 3.68	28.39 ± 1.68	28.76 ± 1.09	27.72 ± 0.47	29.18 ± 1.23	31.36 ± 2.75	30.11 ± 0.56	32.56 ± 1.77	72.05 ± 16.35	29.18 ± 0.68	28.58 ± 0.85	48.65 ± 5.52	29.01 ± 0.87	27.98 ± 0.31	*26.45 ± 0.78*	26.89 ± 0.34	**24.90 ± 0.30**
	6	33.72 ± 1.32	28.96 ± 0.54	32.19 ± 1.49	31.48 ± 0.87	56.34 ± 3.57	29.38 ± 1.65	29.68 ± 1.25	28.49 ± 0.42	29.83 ± 1.43	32.15 ± 2.73	30.93 ± 0.66	34.45 ± 0.97	71.50 ± 17.36	30.20 ± 0.54	29.77 ± 0.83	50.63 ± 5.68	30.06 ± 0.84	30.05 ± 0.16	*26.95 ± 0.72*	27.82 ± 0.31	**25.43 ± 0.28**
	12	36.11 ± 1.21	33.01 ± 0.67	35.35 ± 1.62	34.48 ± 1.00	55.71 ± 2.61	32.06 ± 1.76	32.23 ± 2.13	30.86 ± 0.53	32.41 ± 2.62	34.43 ± 2.95	33.64 ± 0.77	39.71 ± 1.33	76.51 ± 21.48	32.95 ± 0.32	32.74 ± 0.84	55.75 ± 6.17	32.67 ± 0.66	35.49 ± 0.49	*28.12 ± 0.60*	30.08 ± 0.29	**26.51 ± 0.23**
	Avg	34.16 ± 1.30	29.41 ± 0.57	32.84 ± 1.46	31.84 ± 0.85	56.61 ± 3.41	29.69 ± 1.68	30.03 ± 1.38	28.81 ± 0.46	30.24 ± 1.66	32.41 ± 2.78	31.32 ± 0.64	35.09 ± 1.00	72.82 ± 17.76	30.52 ± 0.50	30.10 ± 0.85	51.26 ± 5.73	30.36 ± 0.79	30.72 ± 0.14	*27.06 ± 0.72*	28.07 ± 0.31	**25.52 ± 0.27**

PEMS04

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	STAEFormer	A2TTA (OL-AN)	A2TTA (STAE)
MAE	3	18.47 ± 0.57	16.74 ± 0.18	19.89 ± 0.22	18.89 ± 0.52	141.85 ± 6.13	16.42 ± 0.18	16.61 ± 0.17	15.91 ± 0.18	15.94 ± 0.22	17.20 ± 2.27	17.38 ± 1.52	28.58 ± 9.26	95.04 ± 23.14	16.42 ± 0.18	18.91 ± 0.78	19.98 ± 0.88	16.38 ± 0.21	17.45 ± 0.12	15.32 ± 0.94	*13.88 ± 0.03*	**13.20 ± 0.12**
	6	18.82 ± 0.50	18.38 ± 0.13	20.63 ± 0.21	19.79 ± 0.48	141.72 ± 5.90	17.67 ± 0.12	17.83 ± 0.25	17.18 ± 0.22	17.19 ± 0.24	18.51 ± 2.21	18.38 ± 1.16	30.20 ± 8.75	93.28 ± 23.79	17.67 ± 0.12	19.77 ± 0.83	21.10 ± 0.79	17.59 ± 0.13	19.07 ± 0.08	15.97 ± 0.95	*15.22 ± 0.05*	**13.99 ± 0.16**
	12	20.98 ± 0.43	22.06 ± 0.13	23.02 ± 0.16	22.69 ± 0.40	141.85 ± 5.93	20.41 ± 0.10	20.42 ± 0.48	19.76 ± 0.26	19.81 ± 0.40	21.36 ± 2.15	21.09 ± 1.05	33.75 ± 7.26	89.86 ± 24.96	20.41 ± 0.10	22.34 ± 0.76	23.85 ± 0.72	20.29 ± 0.13	22.95 ± 0.07	*17.15 ± 1.03*	17.67 ± 0.10	**15.28 ± 0.26**
	Avg	19.26 ± 0.51	18.75 ± 0.14	21.00 ± 0.20	20.21 ± 0.47	141.81 ± 5.94	17.93 ± 0.14	18.06 ± 0.29	17.39 ± 0.22	17.42 ± 0.27	18.77 ± 2.21	18.73 ± 1.26	30.55 ± 8.50	92.97 ± 23.88	17.93 ± 0.14	20.12 ± 0.79	21.42 ± 0.79	17.85 ± 0.16	19.50 ± 0.09	16.04 ± 0.96	*15.36 ± 0.06*	**14.02 ± 0.17**
RMSE	3	28.52 ± 0.81	26.47 ± 0.33	33.20 ± 0.42	28.70 ± 0.50	175.11 ± 7.33	25.67 ± 0.19	25.87 ± 0.18	25.10 ± 0.17	25.16 ± 0.21	26.42 ± 2.38	26.66 ± 1.54	40.92 ± 11.22	121.69 ± 26.31	25.67 ± 0.19	30.74 ± 1.80	31.19 ± 1.36	25.64 ± 0.22	27.28 ± 0.09	24.39 ± 1.10	*23.12 ± 0.05*	**22.14 ± 0.15**
	6	29.57 ± 0.72	29.42 ± 0.35	34.61 ± 0.39	30.42 ± 0.50	175.28 ± 7.02	27.99 ± 0.13	28.12 ± 0.22	27.39 ± 0.23	27.45 ± 0.25	28.76 ± 2.26	28.68 ± 1.13	43.56 ± 10.33	120.13 ± 26.82	27.99 ± 0.13	32.27 ± 1.81	32.95 ± 1.27	27.85 ± 0.13	30.32 ± 0.07	25.72 ± 1.12	*25.41 ± 0.08*	**23.60 ± 0.21**
	12	33.41 ± 0.66	35.59 ± 0.48	38.45 ± 0.33	35.28 ± 0.48	176.46 ± 7.19	32.73 ± 0.18	32.48 ± 0.59	31.73 ± 0.33	31.89 ± 0.52	33.56 ± 2.19	33.48 ± 1.20	49.22 ± 7.80	117.63 ± 28.05	32.73 ± 0.18	36.27 ± 1.66	37.15 ± 1.23	32.45 ± 0.17	36.85 ± 0.06	*27.86 ± 1.27*	29.33 ± 0.13	**25.80 ± 0.37**
	Avg	30.18 ± 0.73	29.96 ± 0.38	35.10 ± 0.39	31.05 ± 0.48	175.54 ± 7.10	28.38 ± 0.16	28.43 ± 0.27	27.68 ± 0.23	27.77 ± 0.30	29.15 ± 2.28	29.20 ± 1.30	44.12 ± 9.92	119.98 ± 26.97	28.38 ± 0.16	32.74 ± 1.77	33.40 ± 1.27	28.24 ± 0.17	30.92 ± 0.07	25.79 ± 1.14	*25.58 ± 0.08*	**23.61 ± 0.23**
MAPE (%)	3	24.41 ± 0.81	20.14 ± 1.11	22.62 ± 0.68	22.05 ± 0.96	85.25 ± 2.52	20.70 ± 0.74	20.46 ± 0.84	18.56 ± 0.40	19.95 ± 0.50	25.05 ± 6.64	23.68 ± 2.25	40.16 ± 18.24	68.75 ± 16.29	20.70 ± 0.74	21.38 ± 1.69	30.83 ± 2.62	21.32 ± 0.92	19.56 ± 0.52	18.50 ± 1.05	*17.68 ± 0.30*	**16.27 ± 0.15**
	6	24.55 ± 0.86	21.68 ± 1.13	22.99 ± 0.66	22.75 ± 0.99	85.09 ± 2.62	21.77 ± 0.66	21.54 ± 1.10	19.57 ± 0.42	20.93 ± 0.59	26.57 ± 6.82	24.21 ± 1.83	41.10 ± 16.72	67.26 ± 14.83	21.77 ± 0.66	22.10 ± 1.64	32.08 ± 2.58	22.18 ± 0.89	21.22 ± 0.59	*18.82 ± 1.12*	18.89 ± 0.42	**16.73 ± 0.20**
	12	26.34 ± 0.90	25.76 ± 1.13	25.51 ± 0.72	25.07 ± 1.00	84.82 ± 2.70	24.66 ± 0.56	24.25 ± 1.77	22.28 ± 0.50	23.77 ± 1.13	29.82 ± 6.76	26.69 ± 1.56	44.40 ± 15.20	65.94 ± 11.79	24.66 ± 0.56	24.59 ± 1.68	35.38 ± 2.69	24.44 ± 0.92	26.09 ± 0.96	*19.86 ± 1.25*	21.43 ± 0.66	**17.82 ± 0.27**
	Avg	24.95 ± 0.85	22.18 ± 1.11	23.51 ± 0.66	23.10 ± 0.97	85.10 ± 2.60	22.13 ± 0.65	21.87 ± 1.18	19.91 ± 0.39	21.32 ± 0.69	26.86 ± 6.67	24.60 ± 1.84	41.48 ± 16.83	67.39 ± 14.30	22.13 ± 0.65	22.49 ± 1.67	32.51 ± 2.61	22.47 ± 0.92	21.90 ± 0.60	*18.97 ± 1.13*	19.11 ± 0.44	**16.85 ± 0.21**

PEMS05

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	STAEFormer	A2TTA (OL-AN)	A2TTA (STAE)
MAE	3	9.93 ± 0.19	9.51 ± 0.06	9.78 ± 0.10	10.31 ± 0.14	22.71 ± 0.93	9.38 ± 0.13	12.48 ± 0.66	9.19 ± 0.08	9.45 ± 0.16	9.45 ± 0.18	9.59 ± 0.17	14.09 ± 3.76	29.26 ± 4.54	9.38 ± 0.13	9.50 ± 0.07	10.56 ± 0.12	9.43 ± 0.13	9.72 ± 0.04	8.68 ± 0.10	*8.40 ± 0.01*	**8.02 ± 0.05**
	6	10.18 ± 0.14	10.24 ± 0.06	10.20 ± 0.09	10.87 ± 0.17	23.09 ± 0.92	9.91 ± 0.12	13.60 ± 0.76	9.70 ± 0.05	10.01 ± 0.16	10.00 ± 0.18	10.13 ± 0.13	14.63 ± 3.89	29.75 ± 4.22	9.91 ± 0.12	9.98 ± 0.06	11.00 ± 0.14	9.96 ± 0.12	10.53 ± 0.03	8.90 ± 0.08	*8.89 ± 0.02*	**8.28 ± 0.05**
	12	11.13 ± 0.13	11.75 ± 0.08	11.27 ± 0.08	12.36 ± 0.24	24.12 ± 1.05	11.05 ± 0.12	15.85 ± 0.91	10.72 ± 0.06	11.18 ± 0.19	11.19 ± 0.17	11.32 ± 0.10	15.86 ± 4.06	31.27 ± 3.26	11.05 ± 0.12	11.06 ± 0.06	12.02 ± 0.22	11.08 ± 0.12	12.32 ± 0.05	*9.29 ± 0.09*	9.84 ± 0.04	**8.73 ± 0.06**
	Avg	10.34 ± 0.15	10.37 ± 0.07	10.32 ± 0.09	11.06 ± 0.18	23.23 ± 0.93	10.01 ± 0.12	13.79 ± 0.75	9.78 ± 0.05	10.11 ± 0.17	10.11 ± 0.18	10.24 ± 0.13	14.75 ± 3.87	29.93 ± 4.01	10.01 ± 0.12	10.08 ± 0.06	11.11 ± 0.15	10.06 ± 0.12	10.70 ± 0.04	*8.92 ± 0.09*	8.95 ± 0.02	**8.30 ± 0.05**
RMSE	3	15.17 ± 0.27	14.74 ± 0.08	15.22 ± 0.16	15.65 ± 0.16	29.31 ± 1.00	14.50 ± 0.17	19.73 ± 0.95	14.22 ± 0.06	14.63 ± 0.19	14.56 ± 0.22	14.81 ± 0.21	19.83 ± 3.64	40.76 ± 5.67	14.50 ± 0.17	14.79 ± 0.11	15.74 ± 0.20	14.55 ± 0.12	15.09 ± 0.04	13.43 ± 0.13	*13.28 ± 0.02*	**12.60 ± 0.08**
	6	15.77 ± 0.20	16.04 ± 0.09	16.07 ± 0.13	16.67 ± 0.20	30.01 ± 1.04	15.54 ± 0.17	21.91 ± 1.10	15.19 ± 0.05	15.69 ± 0.17	15.64 ± 0.22	15.88 ± 0.14	20.77 ± 3.82	42.02 ± 5.21	15.54 ± 0.17	15.64 ± 0.09	16.48 ± 0.21	15.55 ± 0.12	16.57 ± 0.03	*13.86 ± 0.11*	14.21 ± 0.04	**13.08 ± 0.09**
	12	17.55 ± 0.18	18.59 ± 0.11	17.99 ± 0.15	19.18 ± 0.34	31.69 ± 1.30	17.59 ± 0.19	26.21 ± 1.34	17.00 ± 0.12	17.83 ± 0.19	17.81 ± 0.24	18.03 ± 0.15	22.71 ± 3.96	45.02 ± 3.88	17.59 ± 0.19	17.46 ± 0.08	18.10 ± 0.35	17.56 ± 0.13	19.59 ± 0.06	*14.58 ± 0.13*	15.86 ± 0.07	**13.87 ± 0.11**
	Avg	16.01 ± 0.22	16.23 ± 0.09	16.26 ± 0.14	16.97 ± 0.22	30.21 ± 1.04	15.69 ± 0.17	22.27 ± 1.07	15.30 ± 0.04	15.86 ± 0.19	15.81 ± 0.23	16.05 ± 0.15	20.92 ± 3.77	42.29 ± 4.93	15.69 ± 0.17	15.79 ± 0.09	16.63 ± 0.22	15.71 ± 0.12	16.82 ± 0.04	*13.89 ± 0.12*	14.29 ± 0.04	**13.10 ± 0.09**
MAPE (%)	3	29.08 ± 1.19	26.80 ± 1.01	29.21 ± 1.30	27.96 ± 1.09	173.89 ± 6.70	27.03 ± 1.26	29.77 ± 0.81	25.17 ± 0.69	26.88 ± 1.51	27.73 ± 1.62	27.83 ± 0.85	71.42 ± 56.45	55.74 ± 4.64	27.03 ± 1.26	26.71 ± 0.44	40.80 ± 1.48	26.58 ± 0.87	27.13 ± 1.01	25.48 ± 0.83	**23.28 ± 0.16**	*23.53 ± 0.26*
	6	29.37 ± 1.05	28.49 ± 1.10	29.88 ± 1.30	29.09 ± 1.04	174.56 ± 7.14	27.88 ± 1.03	31.17 ± 1.05	25.95 ± 0.58	27.80 ± 1.76	28.56 ± 1.52	28.91 ± 0.85	71.92 ± 56.18	56.24 ± 3.25	27.88 ± 1.03	27.90 ± 0.63	42.33 ± 1.80	27.60 ± 0.87	29.07 ± 1.00	26.00 ± 0.67	*24.11 ± 0.17*	**24.05 ± 0.24**
	12	31.14 ± 0.96	32.45 ± 1.23	32.31 ± 1.18	32.26 ± 0.96	177.73 ± 9.82	30.18 ± 0.93	34.34 ± 1.63	28.02 ± 0.40	30.11 ± 2.34	30.89 ± 1.32	31.77 ± 0.88	75.87 ± 55.93	59.82 ± 4.82	30.18 ± 0.93	30.75 ± 0.81	46.57 ± 2.72	30.08 ± 0.84	34.24 ± 1.03	27.46 ± 0.72	*25.90 ± 0.18*	**25.21 ± 0.24**
	Avg	29.71 ± 1.07	28.88 ± 1.09	30.26 ± 1.27	29.50 ± 1.03	175.25 ± 7.82	28.15 ± 1.07	31.49 ± 1.11	26.18 ± 0.54	28.07 ± 1.83	28.85 ± 1.51	29.23 ± 0.88	72.67 ± 56.13	56.93 ± 3.21	28.15 ± 1.07	28.19 ± 0.60	42.89 ± 1.88	27.87 ± 0.85	29.70 ± 1.01	26.20 ± 0.78	*24.27 ± 0.16*	**24.16 ± 0.25**

PEMS06

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	STAEFormer	A2TTA (OL-AN)	A2TTA (STAE)
MAE	3	12.26 ± 0.14	11.99 ± 0.05	12.19 ± 0.04	12.69 ± 0.29	73.37 ± 1.26	11.48 ± 0.03	16.35 ± 0.20	11.20 ± 0.15	11.41 ± 0.10	12.18 ± 1.14	11.86 ± 0.15	16.49 ± 3.90	49.62 ± 9.33	11.48 ± 0.03	11.87 ± 0.16	13.45 ± 0.26	11.77 ± 0.12	12.24 ± 0.08	*10.29 ± 0.07*	10.50 ± 0.06	**9.80 ± 0.06**
	6	12.53 ± 0.09	12.92 ± 0.08	12.66 ± 0.03	13.42 ± 0.26	73.89 ± 1.25	12.14 ± 0.04	17.06 ± 0.36	11.85 ± 0.20	12.07 ± 0.13	12.82 ± 1.12	12.51 ± 0.15	17.64 ± 3.66	49.03 ± 8.94	12.14 ± 0.04	12.45 ± 0.12	14.08 ± 0.17	12.44 ± 0.13	13.29 ± 0.06	*10.55 ± 0.07*	11.10 ± 0.09	**10.08 ± 0.07**
	12	13.73 ± 0.10	14.87 ± 0.15	13.92 ± 0.04	15.17 ± 0.27	74.32 ± 1.51	13.55 ± 0.06	18.56 ± 0.71	13.16 ± 0.30	13.47 ± 0.18	14.22 ± 1.12	14.00 ± 0.18	20.05 ± 3.23	47.02 ± 8.81	13.55 ± 0.06	13.85 ± 0.11	15.47 ± 0.12	13.93 ± 0.18	15.57 ± 0.04	*11.03 ± 0.09*	12.20 ± 0.13	**10.56 ± 0.09**
	Avg	12.74 ± 0.11	13.08 ± 0.09	12.81 ± 0.03	13.61 ± 0.27	73.78 ± 1.30	12.26 ± 0.04	17.20 ± 0.38	11.95 ± 0.21	12.18 ± 0.13	12.94 ± 1.12	12.65 ± 0.16	17.85 ± 3.62	48.64 ± 8.99	12.26 ± 0.04	12.60 ± 0.13	14.22 ± 0.18	12.58 ± 0.14	13.50 ± 0.06	*10.58 ± 0.08*	11.16 ± 0.09	**10.09 ± 0.07**
RMSE	3	18.49 ± 0.20	18.41 ± 0.08	19.04 ± 0.07	19.20 ± 0.49	90.13 ± 1.89	17.47 ± 0.01	23.67 ± 0.41	16.97 ± 0.28	17.30 ± 0.17	18.25 ± 1.36	17.91 ± 0.17	23.98 ± 5.19	64.41 ± 12.49	17.47 ± 0.01	18.38 ± 0.38	19.83 ± 0.41	17.86 ± 0.14	18.87 ± 0.09	*15.53 ± 0.12*	15.94 ± 0.13	**14.77 ± 0.10**
	6	19.09 ± 0.13	20.13 ± 0.13	19.95 ± 0.10	20.55 ± 0.42	91.22 ± 1.84	18.74 ± 0.05	25.10 ± 0.79	18.16 ± 0.38	18.57 ± 0.21	19.52 ± 1.31	19.20 ± 0.21	26.08 ± 4.68	64.32 ± 11.35	18.74 ± 0.05	19.37 ± 0.31	20.88 ± 0.32	19.12 ± 0.16	20.88 ± 0.08	*16.04 ± 0.13*	17.05 ± 0.18	**15.30 ± 0.11**
	12	21.36 ± 0.16	23.44 ± 0.24	22.23 ± 0.12	23.58 ± 0.43	92.76 ± 2.10	21.21 ± 0.11	27.86 ± 1.56	20.43 ± 0.51	20.99 ± 0.30	22.01 ± 1.31	21.85 ± 0.28	30.31 ± 3.77	63.18 ± 10.37	21.21 ± 0.11	21.69 ± 0.31	23.10 ± 0.30	21.70 ± 0.22	24.85 ± 0.09	*16.89 ± 0.19*	18.99 ± 0.25	**16.14 ± 0.17**
	Avg	19.46 ± 0.16	20.34 ± 0.14	20.21 ± 0.09	20.85 ± 0.44	91.16 ± 1.89	18.90 ± 0.05	25.30 ± 0.83	18.29 ± 0.37	18.71 ± 0.22	19.68 ± 1.33	19.38 ± 0.21	26.42 ± 4.59	63.95 ± 11.41	18.90 ± 0.05	19.60 ± 0.33	21.08 ± 0.34	19.32 ± 0.16	21.17 ± 0.09	*16.06 ± 0.14*	17.13 ± 0.17	**15.31 ± 0.12**
MAPE (%)	3	23.42 ± 1.05	20.99 ± 0.64	22.47 ± 0.55	22.62 ± 0.99	83.70 ± 1.48	21.26 ± 0.42	24.94 ± 0.54	20.40 ± 0.70	20.94 ± 0.50	24.69 ± 4.90	23.45 ± 0.81	34.25 ± 15.79	62.73 ± 3.62	21.26 ± 0.42	20.41 ± 0.45	30.46 ± 0.93	22.16 ± 1.38	21.46 ± 0.43	18.75 ± 0.43	*18.72 ± 0.29*	**17.64 ± 0.21**
	6	23.69 ± 0.98	22.28 ± 0.75	23.05 ± 0.55	23.58 ± 1.05	83.77 ± 1.69	22.09 ± 0.30	25.80 ± 0.64	21.26 ± 0.73	21.70 ± 0.44	25.49 ± 4.84	24.14 ± 0.79	35.55 ± 15.53	62.93 ± 4.12	22.09 ± 0.30	21.25 ± 0.35	31.56 ± 0.92	23.16 ± 1.32	23.15 ± 0.50	*19.06 ± 0.45*	19.44 ± 0.29	**18.00 ± 0.19**
	12	25.28 ± 0.99	25.44 ± 0.99	25.06 ± 0.45	26.08 ± 1.17	83.24 ± 1.63	24.24 ± 0.14	27.69 ± 1.04	23.14 ± 1.10	23.52 ± 0.46	27.47 ± 4.93	26.04 ± 0.92	37.91 ± 15.25	62.94 ± 7.33	24.24 ± 0.14	23.34 ± 0.31	34.30 ± 1.11	25.42 ± 1.37	27.28 ± 0.71	*19.77 ± 0.48*	20.98 ± 0.39	**18.77 ± 0.17**
	Avg	23.99 ± 1.00	22.62 ± 0.77	23.35 ± 0.52	23.87 ± 1.07	83.65 ± 1.57	22.33 ± 0.28	25.98 ± 0.69	21.44 ± 0.81	21.89 ± 0.44	25.69 ± 4.86	24.35 ± 0.85	35.70 ± 15.61	62.86 ± 4.31	22.33 ± 0.28	21.49 ± 0.35	31.88 ± 0.95	23.37 ± 1.36	23.62 ± 0.52	*19.13 ± 0.46*	19.58 ± 0.30	**18.06 ± 0.19**

PEMS07

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	STAEFormer	A2TTA (OL-AN)	A2TTA (STAE)
MAE	3	17.68 ± 0.37	15.28 ± 0.18	21.13 ± 0.39	16.99 ± 1.06	40.74 ± 2.00	14.61 ± 0.15	13.99 ± 0.17	13.99 ± 0.21	13.83 ± 0.09	14.90 ± 0.78	13.90 ± 0.16	15.31 ± 0.41	43.00 ± 17.40	14.61 ± 0.15	15.71 ± 0.95	21.00 ± 0.76	14.87 ± 0.09	15.15 ± 0.03	13.28 ± 0.36	*12.75 ± 0.06*	**11.97 ± 0.08**
	6	17.85 ± 0.33	17.00 ± 0.17	21.67 ± 0.38	17.87 ± 0.90	40.91 ± 1.72	15.96 ± 0.10	15.25 ± 0.13	15.41 ± 0.21	15.18 ± 0.11	16.34 ± 0.75	15.30 ± 0.15	16.26 ± 0.33	46.99 ± 19.22	15.96 ± 0.10	16.65 ± 0.83	22.77 ± 0.85	16.23 ± 0.08	17.06 ± 0.04	*13.95 ± 0.30*	14.19 ± 0.09	**12.76 ± 0.08**
	12	19.95 ± 0.32	20.84 ± 0.19	23.86 ± 0.37	20.35 ± 0.89	41.14 ± 1.78	18.69 ± 0.08	17.58 ± 0.12	18.03 ± 0.20	17.67 ± 0.18	18.96 ± 0.73	17.90 ± 0.12	18.22 ± 0.38	57.95 ± 26.22	18.69 ± 0.08	19.27 ± 0.83	26.82 ± 1.11	19.06 ± 0.11	21.20 ± 0.04	*15.05 ± 0.26*	16.68 ± 0.11	**13.93 ± 0.12**
	Avg	18.33 ± 0.34	17.37 ± 0.18	22.05 ± 0.38	18.19 ± 0.94	40.88 ± 1.83	16.16 ± 0.10	15.38 ± 0.13	15.56 ± 0.20	15.33 ± 0.11	16.48 ± 0.76	15.45 ± 0.15	16.40 ± 0.36	48.52 ± 20.66	16.16 ± 0.10	16.98 ± 0.86	23.21 ± 0.86	16.46 ± 0.08	17.43 ± 0.03	*13.98 ± 0.31*	14.29 ± 0.08	**12.75 ± 0.09**
RMSE	3	29.49 ± 0.73	25.28 ± 0.27	40.81 ± 0.83	27.81 ± 1.57	51.49 ± 1.87	24.05 ± 0.16	23.28 ± 0.19	23.32 ± 0.23	23.08 ± 0.10	24.16 ± 0.83	23.24 ± 0.18	24.79 ± 0.52	61.98 ± 24.22	24.05 ± 0.16	26.38 ± 2.15	33.31 ± 0.95	24.55 ± 0.13	25.39 ± 0.03	23.00 ± 0.64	*22.01 ± 0.06*	**20.99 ± 0.14**
	6	30.19 ± 0.71	28.49 ± 0.26	41.72 ± 0.83	29.35 ± 1.24	52.68 ± 1.50	26.61 ± 0.11	25.59 ± 0.14	25.87 ± 0.24	25.56 ± 0.15	26.73 ± 0.78	25.79 ± 0.16	26.66 ± 0.39	69.26 ± 27.45	26.61 ± 0.11	27.98 ± 1.99	36.13 ± 0.91	27.03 ± 0.09	29.02 ± 0.06	*24.33 ± 0.55*	24.48 ± 0.09	**22.45 ± 0.15**
	12	34.15 ± 0.71	35.20 ± 0.31	44.97 ± 0.83	33.47 ± 1.23	54.52 ± 1.51	31.39 ± 0.11	29.66 ± 0.12	30.30 ± 0.23	29.84 ± 0.23	31.22 ± 0.76	30.21 ± 0.13	30.01 ± 0.51	87.60 ± 38.99	31.39 ± 0.11	32.17 ± 2.00	42.87 ± 1.12	31.84 ± 0.11	36.36 ± 0.08	*26.41 ± 0.50*	28.59 ± 0.12	**24.56 ± 0.24**
	Avg	30.93 ± 0.72	29.04 ± 0.27	42.22 ± 0.83	29.83 ± 1.32	52.69 ± 1.61	26.89 ± 0.12	25.76 ± 0.15	26.06 ± 0.22	25.74 ± 0.15	26.93 ± 0.80	25.97 ± 0.16	26.81 ± 0.45	71.60 ± 29.71	26.89 ± 0.12	28.44 ± 2.04	36.89 ± 0.88	27.35 ± 0.10	29.58 ± 0.05	*24.36 ± 0.57*	24.60 ± 0.08	**22.41 ± 0.16**
MAPE (%)	3	21.78 ± 0.65	18.60 ± 0.23	22.16 ± 0.55	20.53 ± 1.38	41.82 ± 1.50	18.95 ± 0.47	20.26 ± 0.37	17.95 ± 0.30	17.78 ± 0.56	21.09 ± 1.43	17.77 ± 0.59	20.99 ± 0.87	57.86 ± 21.96	18.95 ± 0.47	23.73 ± 2.41	39.51 ± 2.62	19.63 ± 0.48	17.93 ± 0.33	17.13 ± 0.52	*16.14 ± 0.34*	**15.45 ± 0.20**
	6	21.89 ± 0.59	20.13 ± 0.23	22.45 ± 0.57	21.36 ± 1.24	41.00 ± 1.53	19.99 ± 0.48	21.20 ± 0.30	18.95 ± 0.31	18.84 ± 0.67	22.35 ± 1.64	18.87 ± 0.47	22.09 ± 1.03	59.98 ± 22.15	19.99 ± 0.48	24.91 ± 2.32	42.31 ± 3.13	20.60 ± 0.46	19.70 ± 0.32	17.56 ± 0.35	*17.35 ± 0.36*	**16.01 ± 0.15**
	12	23.70 ± 0.55	23.96 ± 0.25	24.73 ± 0.62	23.83 ± 1.17	39.94 ± 2.09	22.37 ± 0.62	23.43 ± 0.59	21.17 ± 0.19	21.09 ± 0.95	24.92 ± 2.08	21.37 ± 0.48	24.61 ± 1.33	66.57 ± 24.29	22.37 ± 0.62	28.46 ± 2.44	48.75 ± 4.32	22.98 ± 0.56	24.20 ± 0.49	*18.55 ± 0.37*	19.77 ± 0.35	**17.07 ± 0.17**
	Avg	22.29 ± 0.60	20.56 ± 0.23	22.92 ± 0.56	21.70 ± 1.27	41.00 ± 1.62	20.23 ± 0.49	21.43 ± 0.30	19.16 ± 0.26	19.04 ± 0.71	22.54 ± 1.66	19.12 ± 0.50	22.35 ± 0.90	60.89 ± 22.44	20.23 ± 0.49	25.41 ± 2.37	43.01 ± 3.23	20.86 ± 0.47	20.22 ± 0.36	17.64 ± 0.38	*17.54 ± 0.35*	**16.07 ± 0.16**

PEMS08

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	STAEFormer	A2TTA (OL-AN)	A2TTA (STAE)
MAE	3	15.09 ± 0.18	13.60 ± 0.05	15.41 ± 0.25	14.69 ± 0.30	55.43 ± 2.75	13.15 ± 0.08	13.23 ± 0.02	12.66 ± 0.10	12.66 ± 0.12	13.52 ± 0.38	13.21 ± 0.47	13.92 ± 0.29	38.58 ± 11.44	13.15 ± 0.08	14.10 ± 0.28	16.83 ± 0.52	13.36 ± 0.10	13.68 ± 0.06	11.91 ± 0.08	*11.77 ± 0.04*	**11.11 ± 0.03**
	6	15.41 ± 0.19	14.99 ± 0.06	16.02 ± 0.20	15.49 ± 0.19	55.14 ± 2.86	14.21 ± 0.06	14.31 ± 0.05	13.75 ± 0.09	13.74 ± 0.13	14.59 ± 0.37	14.19 ± 0.36	14.71 ± 0.28	39.50 ± 10.09	14.21 ± 0.06	14.85 ± 0.25	17.80 ± 0.44	14.43 ± 0.09	15.20 ± 0.03	*12.46 ± 0.07*	12.83 ± 0.04	**11.71 ± 0.05**
	12	17.14 ± 0.19	17.95 ± 0.07	17.84 ± 0.18	17.72 ± 0.13	54.80 ± 3.34	16.32 ± 0.06	16.31 ± 0.11	15.72 ± 0.10	15.70 ± 0.16	16.57 ± 0.35	16.13 ± 0.34	16.26 ± 0.23	42.82 ± 9.63	16.32 ± 0.06	16.84 ± 0.25	20.12 ± 0.36	16.52 ± 0.11	18.47 ± 0.05	*13.40 ± 0.09*	14.66 ± 0.07	**12.68 ± 0.08**
	Avg	15.74 ± 0.18	15.25 ± 0.05	16.27 ± 0.21	15.78 ± 0.20	55.14 ± 2.96	14.37 ± 0.06	14.44 ± 0.04	13.86 ± 0.09	13.85 ± 0.14	14.70 ± 0.37	14.33 ± 0.39	14.80 ± 0.27	40.06 ± 10.41	14.37 ± 0.06	15.09 ± 0.26	18.07 ± 0.43	14.58 ± 0.10	15.49 ± 0.04	*12.50 ± 0.08*	12.91 ± 0.05	**11.73 ± 0.05**
RMSE	3	24.19 ± 0.43	22.34 ± 0.08	27.60 ± 0.56	23.19 ± 0.39	64.45 ± 3.18	21.49 ± 0.09	21.49 ± 0.09	20.91 ± 0.11	20.88 ± 0.14	21.71 ± 0.36	21.48 ± 0.56	22.09 ± 0.33	52.38 ± 15.01	21.49 ± 0.09	24.15 ± 0.97	25.52 ± 0.56	21.76 ± 0.13	22.54 ± 0.07	*19.82 ± 0.06*	19.86 ± 0.04	**18.75 ± 0.06**
	6	25.28 ± 0.43	24.97 ± 0.09	28.91 ± 0.50	24.87 ± 0.24	64.82 ± 3.25	23.69 ± 0.08	23.61 ± 0.10	23.07 ± 0.11	23.06 ± 0.17	23.86 ± 0.32	23.54 ± 0.40	23.74 ± 0.32	54.74 ± 13.61	23.69 ± 0.08	25.56 ± 0.92	27.15 ± 0.43	23.83 ± 0.14	25.49 ± 0.05	*20.99 ± 0.09*	21.95 ± 0.06	**20.02 ± 0.10**
	12	28.59 ± 0.39	30.05 ± 0.12	32.03 ± 0.48	28.77 ± 0.29	65.72 ± 3.74	27.64 ± 0.10	27.28 ± 0.15	26.65 ± 0.13	26.68 ± 0.26	27.52 ± 0.24	27.20 ± 0.52	26.55 ± 0.38	61.06 ± 13.36	27.64 ± 0.10	28.84 ± 0.89	30.90 ± 0.37	27.55 ± 0.18	31.18 ± 0.07	*22.86 ± 0.18*	25.20 ± 0.10	**21.90 ± 0.20**
	Avg	25.73 ± 0.42	25.32 ± 0.09	29.24 ± 0.52	25.27 ± 0.27	64.90 ± 3.37	23.91 ± 0.08	23.78 ± 0.09	23.20 ± 0.11	23.20 ± 0.18	24.01 ± 0.31	23.74 ± 0.49	23.84 ± 0.34	55.56 ± 13.95	23.91 ± 0.08	25.88 ± 0.93	27.54 ± 0.43	24.04 ± 0.14	25.88 ± 0.05	*21.05 ± 0.11*	22.01 ± 0.06	**20.02 ± 0.11**
MAPE (%)	3	23.28 ± 0.94	18.52 ± 0.26	20.74 ± 0.82	20.41 ± 1.14	58.28 ± 1.31	19.47 ± 0.67	19.37 ± 0.37	17.67 ± 0.36	18.93 ± 0.85	22.36 ± 2.84	20.22 ± 1.84	25.77 ± 1.99	54.65 ± 12.58	19.47 ± 0.67	19.88 ± 1.02	37.93 ± 3.07	20.79 ± 1.06	18.72 ± 0.43	16.59 ± 0.37	*15.84 ± 0.11*	**15.50 ± 0.09**
	6	23.45 ± 0.90	19.97 ± 0.29	21.27 ± 0.87	21.21 ± 1.14	57.76 ± 1.83	20.60 ± 0.53	20.50 ± 0.40	18.75 ± 0.37	20.04 ± 0.83	23.80 ± 3.22	21.20 ± 1.64	25.87 ± 1.74	57.11 ± 13.23	20.60 ± 0.53	20.62 ± 0.94	39.63 ± 2.92	21.87 ± 1.02	20.43 ± 0.37	17.01 ± 0.38	*16.85 ± 0.10*	**15.95 ± 0.09**
	12	25.07 ± 0.83	23.50 ± 0.43	23.80 ± 0.88	23.76 ± 1.19	57.53 ± 2.82	23.17 ± 0.38	22.98 ± 0.65	21.29 ± 0.43	22.60 ± 0.84	26.73 ± 3.92	23.46 ± 1.08	27.18 ± 1.94	63.81 ± 14.26	23.17 ± 0.38	23.08 ± 0.84	43.69 ± 2.80	24.31 ± 0.92	24.67 ± 0.42	*18.04 ± 0.37*	19.03 ± 0.14	**16.91 ± 0.14**
	Avg	23.79 ± 0.89	20.35 ± 0.31	21.74 ± 0.85	21.58 ± 1.15	57.94 ± 1.86	20.86 ± 0.52	20.74 ± 0.45	19.03 ± 0.36	20.30 ± 0.81	24.03 ± 3.24	21.44 ± 1.53	26.11 ± 1.53	58.03 ± 13.17	20.86 ± 0.52	20.98 ± 0.93	40.10 ± 2.92	22.13 ± 1.00	20.91 ± 0.38	17.14 ± 0.36	*17.06 ± 0.11*	**16.04 ± 0.10**

PEMS10

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	STAEFormer	A2TTA (OL-AN)	A2TTA (STAE)
MAE	3	9.61 ± 0.10	9.23 ± 0.04	9.64 ± 0.08	9.65 ± 0.13	85.24 ± 3.68	8.97 ± 0.06	9.01 ± 0.18	8.84 ± 0.04	9.00 ± 0.07	9.10 ± 0.28	9.01 ± 0.05	10.94 ± 2.01	24.47 ± 4.51	8.97 ± 0.06	9.58 ± 0.06	10.66 ± 0.21	9.01 ± 0.05	9.40 ± 0.04	*8.23 ± 0.04*	8.40 ± 0.02	**7.88 ± 0.03**
	6	9.85 ± 0.09	9.92 ± 0.03	10.02 ± 0.06	10.07 ± 0.13	85.46 ± 3.75	9.48 ± 0.06	9.50 ± 0.20	9.36 ± 0.06	9.53 ± 0.08	9.62 ± 0.25	9.51 ± 0.06	11.43 ± 2.05	24.49 ± 4.89	9.48 ± 0.06	10.02 ± 0.05	11.07 ± 0.23	9.50 ± 0.06	10.14 ± 0.04	*8.42 ± 0.03*	8.88 ± 0.03	**8.07 ± 0.03**
	12	10.78 ± 0.10	11.36 ± 0.02	10.98 ± 0.07	11.16 ± 0.16	85.57 ± 3.77	10.56 ± 0.04	10.54 ± 0.27	10.40 ± 0.08	10.64 ± 0.11	10.72 ± 0.23	10.56 ± 0.08	12.59 ± 2.24	27.60 ± 10.02	10.56 ± 0.04	11.15 ± 0.08	12.05 ± 0.28	10.51 ± 0.06	11.76 ± 0.05	*8.75 ± 0.03*	9.78 ± 0.05	**8.41 ± 0.03**
	Avg	10.01 ± 0.10	10.05 ± 0.03	10.13 ± 0.07	10.21 ± 0.13	85.42 ± 3.64	9.58 ± 0.05	9.60 ± 0.21	9.45 ± 0.05	9.63 ± 0.09	9.72 ± 0.25	9.60 ± 0.06	11.55 ± 2.08	25.28 ± 6.03	9.58 ± 0.05	10.15 ± 0.06	11.18 ± 0.23	9.59 ± 0.06	10.30 ± 0.04	*8.44 ± 0.03*	8.94 ± 0.03	**8.08 ± 0.03**
RMSE	3	15.54 ± 0.18	15.15 ± 0.09	16.68 ± 0.12	15.58 ± 0.15	87.14 ± 3.76	14.56 ± 0.07	14.67 ± 0.38	14.33 ± 0.09	14.61 ± 0.07	14.79 ± 0.40	14.65 ± 0.07	16.33 ± 1.74	35.53 ± 7.61	14.56 ± 0.07	15.51 ± 0.08	16.76 ± 0.35	14.71 ± 0.08	15.37 ± 0.06	*13.36 ± 0.07*	13.74 ± 0.05	**12.84 ± 0.06**
	6	16.11 ± 0.17	16.47 ± 0.06	17.39 ± 0.11	16.43 ± 0.16	87.70 ± 3.90	15.62 ± 0.07	15.71 ± 0.43	15.40 ± 0.12	15.67 ± 0.12	15.86 ± 0.34	15.70 ± 0.07	17.34 ± 1.73	36.35 ± 8.62	15.62 ± 0.07	16.36 ± 0.07	17.50 ± 0.42	15.72 ± 0.09	16.82 ± 0.05	*13.76 ± 0.06*	14.70 ± 0.08	**13.26 ± 0.06**
	12	17.95 ± 0.18	19.07 ± 0.04	19.14 ± 0.10	18.42 ± 0.24	88.47 ± 4.01	17.71 ± 0.04	17.71 ± 0.51	17.43 ± 0.14	17.76 ± 0.21	17.95 ± 0.27	17.75 ± 0.11	19.33 ± 1.87	42.28 ± 17.28	17.71 ± 0.04	18.34 ± 0.13	19.22 ± 0.53	17.63 ± 0.09	19.73 ± 0.05	*14.44 ± 0.05*	16.47 ± 0.11	**13.93 ± 0.05**
	Avg	16.39 ± 0.17	16.67 ± 0.06	17.59 ± 0.11	16.64 ± 0.17	87.72 ± 3.79	15.78 ± 0.06	15.85 ± 0.43	15.54 ± 0.11	15.84 ± 0.12	16.02 ± 0.34	15.85 ± 0.07	17.49 ± 1.75	37.58 ± 10.52	15.78 ± 0.06	16.56 ± 0.09	17.68 ± 0.41	15.85 ± 0.08	17.05 ± 0.05	*13.80 ± 0.06*	14.81 ± 0.07	**13.27 ± 0.06**
MAPE (%)	3	34.43 ± 0.71	30.69 ± 0.68	32.80 ± 0.59	32.67 ± 1.20	991.29 ± 37.75	32.60 ± 0.56	33.04 ± 1.77	31.82 ± 0.52	33.69 ± 1.03	32.26 ± 1.52	32.98 ± 0.77	71.76 ± 52.11	141.42 ± 36.47	32.60 ± 0.56	35.57 ± 0.64	45.20 ± 1.86	32.03 ± 0.84	31.54 ± 0.83	30.26 ± 0.79	*29.37 ± 0.14*	**28.23 ± 0.13**
	6	35.19 ± 0.68	32.34 ± 0.74	33.68 ± 0.57	33.64 ± 1.15	988.78 ± 36.98	33.69 ± 0.47	34.17 ± 1.50	32.68 ± 0.39	35.07 ± 0.85	33.34 ± 1.55	34.10 ± 0.89	72.30 ± 53.03	134.84 ± 33.38	33.69 ± 0.47	36.84 ± 0.69	46.38 ± 1.92	33.14 ± 1.03	33.60 ± 0.82	30.62 ± 0.85	*30.15 ± 0.12*	**28.63 ± 0.17**
	12	37.83 ± 0.85	36.47 ± 0.85	36.56 ± 0.56	36.59 ± 1.11	983.86 ± 32.30	36.64 ± 0.37	37.38 ± 2.07	35.61 ± 0.44	39.28 ± 0.97	36.40 ± 1.71	36.83 ± 0.88	76.77 ± 55.91	136.12 ± 31.89	36.64 ± 0.37	40.42 ± 0.74	49.58 ± 2.08	35.90 ± 1.40	38.86 ± 0.76	*31.66 ± 0.88*	32.16 ± 0.14	**29.59 ± 0.19**
	Avg	35.60 ± 0.74	32.82 ± 0.73	34.11 ± 0.55	34.06 ± 1.15	988.49 ± 35.46	34.07 ± 0.45	34.63 ± 1.67	33.14 ± 0.41	35.68 ± 0.83	33.74 ± 1.57	34.43 ± 0.84	73.27 ± 53.44	137.49 ± 33.01	34.07 ± 0.45	37.30 ± 0.68	46.79 ± 1.93	33.45 ± 1.06	34.23 ± 0.81	30.76 ± 0.86	*30.39 ± 0.12*	**28.74 ± 0.16**

PEMS11

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	STAEFormer	A2TTA (OL-AN)	A2TTA (STAE)
MAE	3	17.96 ± 0.26	16.26 ± 0.08	19.33 ± 0.31	17.66 ± 0.48	89.03 ± 3.90	15.63 ± 0.11	18.13 ± 1.34	15.18 ± 0.20	15.52 ± 0.21	18.89 ± 1.27	16.10 ± 0.27	22.68 ± 3.86	56.58 ± 9.83	15.63 ± 0.11	16.96 ± 0.51	19.41 ± 0.47	15.78 ± 0.18	16.53 ± 0.16	15.04 ± 0.38	*13.88 ± 0.07*	**13.26 ± 0.10**
	6	18.40 ± 0.16	18.04 ± 0.06	19.98 ± 0.26	18.63 ± 0.43	90.62 ± 3.59	17.09 ± 0.07	19.57 ± 1.37	16.66 ± 0.17	17.04 ± 0.19	20.26 ± 1.21	17.61 ± 0.20	24.20 ± 4.29	58.08 ± 9.18	17.09 ± 0.07	17.91 ± 0.43	20.77 ± 0.61	17.22 ± 0.16	18.37 ± 0.11	15.66 ± 0.34	*15.35 ± 0.08*	**14.14 ± 0.12**
	12	20.91 ± 0.14	21.91 ± 0.11	22.44 ± 0.22	21.39 ± 0.63	94.00 ± 3.26	20.15 ± 0.09	22.64 ± 1.45	19.55 ± 0.15	20.07 ± 0.26	23.09 ± 1.18	20.66 ± 0.19	28.03 ± 4.78	62.80 ± 7.57	20.15 ± 0.09	20.61 ± 0.44	23.74 ± 0.97	20.31 ± 0.17	22.56 ± 0.10	*16.96 ± 0.34*	18.06 ± 0.11	**15.70 ± 0.17**
	Avg	18.89 ± 0.19	18.40 ± 0.06	20.39 ± 0.26	18.99 ± 0.50	90.91 ± 3.59	17.35 ± 0.08	19.85 ± 1.35	16.87 ± 0.17	17.28 ± 0.21	20.49 ± 1.22	17.85 ± 0.21	24.66 ± 4.24	58.83 ± 8.65	17.35 ± 0.08	18.26 ± 0.45	21.07 ± 0.64	17.49 ± 0.17	18.79 ± 0.13	15.77 ± 0.35	*15.52 ± 0.09*	**14.21 ± 0.12**
RMSE	3	28.48 ± 0.32	26.41 ± 0.18	34.68 ± 0.52	28.43 ± 0.77	114.27 ± 3.60	25.19 ± 0.11	28.88 ± 2.36	24.57 ± 0.19	24.94 ± 0.31	30.78 ± 2.30	25.70 ± 0.24	34.39 ± 4.99	76.83 ± 11.66	25.19 ± 0.11	27.09 ± 1.17	30.55 ± 0.66	25.31 ± 0.25	26.87 ± 0.15	24.09 ± 0.45	*22.97 ± 0.14*	**21.86 ± 0.20**
	6	29.86 ± 0.18	30.02 ± 0.19	36.03 ± 0.45	30.46 ± 0.73	117.21 ± 3.30	28.17 ± 0.07	31.74 ± 2.27	27.50 ± 0.18	27.98 ± 0.27	33.33 ± 2.04	28.78 ± 0.17	37.30 ± 5.62	79.99 ± 10.80	28.17 ± 0.07	28.90 ± 1.25	32.96 ± 0.95	28.21 ± 0.22	30.66 ± 0.10	*25.51 ± 0.38*	25.91 ± 0.14	**23.78 ± 0.22**
	12	35.01 ± 0.18	37.34 ± 0.31	40.33 ± 0.38	35.57 ± 1.14	123.42 ± 2.95	34.02 ± 0.13	37.46 ± 2.32	32.97 ± 0.17	33.67 ± 0.38	38.38 ± 1.64	34.54 ± 0.21	43.91 ± 6.17	88.30 ± 9.44	34.02 ± 0.13	33.63 ± 1.29	38.07 ± 1.64	34.05 ± 0.23	38.51 ± 0.12	*28.23 ± 0.44*	31.07 ± 0.21	**27.00 ± 0.31**
	Avg	30.69 ± 0.22	30.60 ± 0.20	36.66 ± 0.45	31.03 ± 0.85	117.77 ± 3.26	28.59 ± 0.08	32.20 ± 2.27	27.85 ± 0.18	28.35 ± 0.31	33.70 ± 2.02	29.15 ± 0.18	37.97 ± 5.48	81.11 ± 10.21	28.59 ± 0.08	29.46 ± 1.21	33.42 ± 1.02	28.66 ± 0.24	31.32 ± 0.12	*25.69 ± 0.41*	26.17 ± 0.16	**23.90 ± 0.24**
MAPE (%)	3	26.88 ± 0.56	21.14 ± 0.47	25.64 ± 1.01	23.31 ± 0.72	75.67 ± 11.80	22.19 ± 0.48	25.16 ± 0.69	20.83 ± 0.76	22.54 ± 0.92	31.45 ± 8.34	24.34 ± 1.56	40.12 ± 12.74	73.77 ± 4.97	22.19 ± 0.48	26.10 ± 0.83	38.80 ± 2.29	23.12 ± 1.18	21.22 ± 0.49	21.83 ± 0.86	*19.95 ± 0.28*	**18.90 ± 0.26**
	6	26.98 ± 0.48	23.01 ± 0.58	25.88 ± 0.76	24.26 ± 0.76	78.26 ± 13.52	23.48 ± 0.40	26.37 ± 0.87	22.08 ± 0.81	23.81 ± 1.02	32.96 ± 8.38	25.54 ± 1.29	41.79 ± 11.65	77.24 ± 6.76	23.48 ± 0.40	27.12 ± 0.75	41.11 ± 2.84	24.34 ± 1.16	23.21 ± 0.41	22.51 ± 0.85	*21.09 ± 0.24*	**19.63 ± 0.32**
	12	29.29 ± 0.34	27.70 ± 0.84	28.71 ± 0.38	27.15 ± 0.82	84.72 ± 17.68	26.75 ± 0.52	29.56 ± 1.23	25.18 ± 0.77	27.18 ± 1.73	36.52 ± 8.49	28.90 ± 0.88	47.15 ± 11.15	87.32 ± 13.09	26.75 ± 0.52	30.71 ± 0.61	46.10 ± 3.30	27.47 ± 1.28	28.54 ± 0.41	24.15 ± 0.94	*23.51 ± 0.26*	**21.14 ± 0.48**
	Avg	27.53 ± 0.47	23.56 ± 0.60	26.51 ± 0.73	24.67 ± 0.76	79.10 ± 13.95	23.86 ± 0.41	26.77 ± 0.88	22.45 ± 0.78	24.25 ± 1.16	33.33 ± 8.37	26.00 ± 1.27	42.58 ± 11.70	78.84 ± 7.49	23.86 ± 0.41	27.69 ± 0.74	41.60 ± 2.73	24.71 ± 1.18	23.89 ± 0.42	22.68 ± 0.85	*21.31 ± 0.25*	**19.75 ± 0.33**

PEMS12

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	STAEFormer	A2TTA (OL-AN)	A2TTA (STAE)
MAE	3	15.00 ± 0.41	13.84 ± 0.11	15.91 ± 0.29	14.75 ± 0.37	15.47 ± 0.26	13.38 ± 0.17	16.88 ± 0.82	12.98 ± 0.08	13.02 ± 0.16	13.69 ± 0.66	12.98 ± 0.11	13.54 ± 0.32	39.70 ± 11.56	13.38 ± 0.17	14.19 ± 0.30	16.38 ± 0.51	13.54 ± 0.32	13.82 ± 0.06	11.95 ± 0.12	*11.30 ± 0.02*	**10.65 ± 0.02**
	6	15.28 ± 0.31	15.18 ± 0.09	16.39 ± 0.26	15.65 ± 0.30	16.55 ± 0.28	14.46 ± 0.14	18.46 ± 0.96	14.11 ± 0.07	14.16 ± 0.12	14.81 ± 0.61	14.09 ± 0.10	14.35 ± 0.30	39.93 ± 10.99	14.46 ± 0.14	14.94 ± 0.20	17.55 ± 0.60	14.66 ± 0.28	15.30 ± 0.04	12.43 ± 0.09	*12.41 ± 0.03*	**11.27 ± 0.03**
	12	17.18 ± 0.24	18.22 ± 0.10	18.25 ± 0.24	17.92 ± 0.25	18.64 ± 0.26	16.72 ± 0.17	21.75 ± 1.22	16.27 ± 0.08	16.36 ± 0.13	17.01 ± 0.54	16.26 ± 0.10	16.13 ± 0.40	41.39 ± 9.57	16.72 ± 0.17	17.19 ± 0.16	20.37 ± 0.81	17.06 ± 0.27	18.58 ± 0.05	*13.27 ± 0.09*	14.38 ± 0.04	**12.21 ± 0.04**
	Avg	15.67 ± 0.33	15.48 ± 0.10	16.70 ± 0.27	15.91 ± 0.30	16.69 ± 0.26	14.65 ± 0.16	18.74 ± 0.97	14.25 ± 0.07	14.31 ± 0.13	14.96 ± 0.61	14.25 ± 0.10	14.51 ± 0.31	40.14 ± 10.64	14.65 ± 0.16	15.24 ± 0.23	17.88 ± 0.60	14.88 ± 0.29	15.61 ± 0.05	*12.47 ± 0.09*	12.51 ± 0.03	**11.28 ± 0.03**
RMSE	3	25.59 ± 0.58	24.01 ± 0.22	29.42 ± 0.56	25.04 ± 0.63	24.85 ± 0.21	22.92 ± 0.24	28.98 ± 1.46	22.39 ± 0.10	22.60 ± 0.19	23.32 ± 0.92	22.57 ± 0.18	22.99 ± 0.32	58.61 ± 16.84	22.92 ± 0.24	24.49 ± 0.41	27.04 ± 0.58	23.25 ± 0.40	24.14 ± 0.05	21.47 ± 0.07	*20.68 ± 0.01*	**19.79 ± 0.07**
	6	26.51 ± 0.44	26.73 ± 0.20	30.47 ± 0.52	26.90 ± 0.46	27.03 ± 0.25	25.16 ± 0.23	32.15 ± 1.73	24.59 ± 0.10	24.87 ± 0.16	25.55 ± 0.86	24.76 ± 0.14	24.71 ± 0.30	59.98 ± 16.14	25.16 ± 0.23	26.00 ± 0.27	29.06 ± 0.68	25.41 ± 0.34	27.22 ± 0.07	*22.63 ± 0.13*	22.80 ± 0.02	**21.11 ± 0.11**
	12	30.21 ± 0.34	32.34 ± 0.25	33.76 ± 0.46	31.14 ± 0.34	30.96 ± 0.27	29.39 ± 0.35	38.59 ± 2.26	28.52 ± 0.13	28.91 ± 0.24	29.64 ± 0.81	28.68 ± 0.18	27.88 ± 0.56	63.86 ± 14.63	29.39 ± 0.35	29.92 ± 0.25	34.01 ± 1.13	29.71 ± 0.31	33.48 ± 0.07	*24.39 ± 0.22*	26.23 ± 0.04	**22.95 ± 0.19**
	Avg	27.11 ± 0.46	27.18 ± 0.22	30.93 ± 0.51	27.32 ± 0.46	27.24 ± 0.23	25.42 ± 0.27	32.67 ± 1.76	24.79 ± 0.10	25.07 ± 0.17	25.78 ± 0.87	24.96 ± 0.14	24.89 ± 0.33	60.33 ± 15.78	25.42 ± 0.27	26.44 ± 0.31	29.63 ± 0.72	25.73 ± 0.35	27.71 ± 0.06	*22.66 ± 0.12*	22.89 ± 0.02	**21.07 ± 0.12**
MAPE (%)	3	27.65 ± 1.25	22.64 ± 0.56	27.12 ± 0.41	24.35 ± 0.86	25.71 ± 0.68	23.51 ± 1.20	25.54 ± 0.20	21.76 ± 0.56	23.13 ± 1.02	26.74 ± 1.11	22.87 ± 0.93	26.37 ± 2.97	88.77 ± 50.47	23.51 ± 1.20	27.44 ± 0.78	44.56 ± 2.81	26.09 ± 2.05	22.81 ± 0.64	21.09 ± 0.73	*18.97 ± 0.17*	**18.80 ± 0.23**
	6	27.81 ± 1.07	24.48 ± 0.58	27.46 ± 0.33	25.49 ± 0.90	26.79 ± 0.78	24.69 ± 1.02	27.05 ± 0.21	22.87 ± 0.53	24.12 ± 0.81	27.91 ± 0.93	24.15 ± 0.65	26.85 ± 2.47	89.73 ± 50.91	24.69 ± 1.02	28.80 ± 0.89	47.26 ± 3.25	27.25 ± 1.76	24.88 ± 0.56	21.54 ± 0.70	*20.26 ± 0.10*	**19.43 ± 0.19**
	12	30.30 ± 0.90	29.30 ± 0.81	30.91 ± 0.40	28.84 ± 1.12	29.29 ± 0.84	27.67 ± 0.81	30.45 ± 0.46	25.78 ± 0.68	26.99 ± 0.51	30.72 ± 1.12	27.24 ± 0.38	29.02 ± 2.20	92.59 ± 50.83	27.67 ± 0.81	33.37 ± 1.06	54.00 ± 4.40	30.22 ± 1.44	30.38 ± 0.72	*22.74 ± 0.62*	22.96 ± 0.15	**20.74 ± 0.17**
	Avg	28.38 ± 1.07	25.06 ± 0.60	28.22 ± 0.29	25.95 ± 0.93	27.05 ± 0.65	25.03 ± 1.03	27.38 ± 0.23	23.21 ± 0.56	24.47 ± 0.78	28.19 ± 1.00	24.49 ± 0.64	27.22 ± 2.49	90.13 ± 50.80	25.03 ± 1.03	29.49 ± 0.89	48.07 ± 3.34	27.62 ± 1.78	25.57 ± 0.60	21.69 ± 0.69	*20.50 ± 0.09*	**19.54 ± 0.20**

TFNSW

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	STAEFormer	A2TTA (OL-AN)	A2TTA (STAE)
MAE	3	154.71 ± 26.20	120.43 ± 2.70	179.02 ± 1.91	151.92 ± 3.36	255.23 ± 13.44	117.50 ± 0.80	143.27 ± 4.50	112.16 ± 1.02	124.40 ± 2.34	153.08 ± 23.52	127.32 ± 0.74	144.99 ± 6.72	206.53 ± 29.83	117.50 ± 0.80	153.59 ± 7.41	194.65 ± 4.95	124.38 ± 1.35	148.93 ± 1.21	117.06 ± 4.49	*76.92 ± 0.35*	**74.81 ± 1.16**
	6	168.09 ± 25.18	142.16 ± 2.15	197.39 ± 2.11	189.66 ± 4.71	257.75 ± 14.37	139.55 ± 1.48	173.40 ± 4.27	132.25 ± 1.60	147.49 ± 2.76	170.96 ± 21.96	150.01 ± 2.00	169.13 ± 6.93	223.59 ± 26.51	139.55 ± 1.48	167.06 ± 5.69	218.88 ± 9.09	144.51 ± 1.43	183.53 ± 1.46	125.79 ± 4.22	*89.87 ± 0.39*	**84.71 ± 1.30**
	12	178.60 ± 24.41	154.85 ± 1.47	207.45 ± 1.81	225.20 ± 5.33	264.38 ± 14.33	153.31 ± 1.53	193.45 ± 4.19	144.90 ± 1.60	161.79 ± 3.13	182.64 ± 20.38	164.66 ± 2.20	188.10 ± 7.48	238.56 ± 17.76	153.31 ± 1.53	182.85 ± 6.17	242.08 ± 5.69	157.40 ± 2.31	196.71 ± 1.43	136.99 ± 3.60	*102.35 ± 0.77*	**94.27 ± 1.23**
	Avg	166.42 ± 25.34	137.44 ± 2.08	194.20 ± 1.94	185.34 ± 4.41	258.78 ± 13.01	135.30 ± 1.27	168.31 ± 3.63	128.04 ± 1.21	143.01 ± 2.76	167.64 ± 21.96	145.72 ± 1.31	165.96 ± 6.53	222.09 ± 23.67	135.30 ± 1.27	167.52 ± 6.16	215.87 ± 6.45	140.64 ± 1.60	174.53 ± 1.27	125.45 ± 4.10	*87.85 ± 0.39*	**83.20 ± 1.15**
RMSE	3	254.27 ± 33.42	210.60 ± 2.82	333.75 ± 4.67	277.08 ± 6.58	361.27 ± 16.80	207.90 ± 1.82	250.70 ± 5.20	199.48 ± 1.74	217.34 ± 3.91	277.36 ± 48.96	217.54 ± 2.03	260.79 ± 14.18	325.86 ± 37.46	207.90 ± 1.82	300.80 ± 12.84	350.20 ± 12.10	219.82 ± 2.30	263.04 ± 1.52	228.93 ± 8.62	**154.43 ± 1.05**	*159.35 ± 2.83*
	6	282.43 ± 31.38	247.96 ± 2.32	361.45 ± 4.47	348.91 ± 8.66	373.62 ± 14.49	249.99 ± 3.15	309.98 ± 6.89	238.37 ± 2.61	262.75 ± 4.67	310.53 ± 44.64	263.30 ± 3.47	310.93 ± 14.83	360.65 ± 25.42	249.99 ± 3.15	331.25 ± 8.50	395.61 ± 19.55	258.97 ± 2.31	319.77 ± 1.61	248.81 ± 7.85	**181.62 ± 1.16**	*182.50 ± 2.81*
	12	301.82 ± 29.86	272.53 ± 1.72	375.70 ± 3.67	402.58 ± 10.07	389.17 ± 12.88	274.52 ± 2.64	343.74 ± 7.94	260.85 ± 2.78	287.91 ± 5.14	329.31 ± 40.17	288.47 ± 3.24	342.93 ± 15.96	384.77 ± 13.77	274.52 ± 2.64	363.91 ± 9.16	435.08 ± 14.01	280.40 ± 2.99	340.07 ± 1.51	272.17 ± 6.32	*209.52 ± 1.71*	**203.77 ± 2.03**
	Avg	277.33 ± 31.79	239.34 ± 2.24	356.00 ± 4.27	336.01 ± 8.45	372.50 ± 13.79	240.06 ± 2.50	297.37 ± 5.65	228.72 ± 2.07	251.94 ± 4.70	302.48 ± 45.06	252.38 ± 2.69	301.10 ± 13.76	354.57 ± 24.38	240.06 ± 2.50	330.41 ± 9.84	388.74 ± 14.74	249.37 ± 2.46	302.47 ± 1.47	247.33 ± 7.59	**177.20 ± 1.17**	*178.38 ± 2.52*
MAPE (%)	3	184.09 ± 66.77	107.26 ± 4.88	157.41 ± 6.71	115.34 ± 7.85	377.66 ± 17.24	98.37 ± 3.94	121.64 ± 17.19	89.56 ± 6.49	93.40 ± 10.17	95.27 ± 12.46	131.58 ± 7.68	112.66 ± 11.25	224.54 ± 97.80	98.37 ± 3.94	105.61 ± 25.71	309.21 ± 51.42	99.42 ± 8.35	156.70 ± 12.90	68.56 ± 3.68	*58.23 ± 2.81*	**56.05 ± 0.94**
	6	192.25 ± 66.30	134.55 ± 5.16	167.88 ± 5.72	148.43 ± 15.69	385.51 ± 34.29	108.31 ± 3.35	133.04 ± 15.87	103.36 ± 5.89	106.28 ± 9.80	108.37 ± 9.61	142.60 ± 10.59	124.52 ± 12.02	246.12 ± 110.14	108.31 ± 3.35	106.49 ± 23.16	337.93 ± 50.78	115.05 ± 14.59	187.96 ± 13.09	73.12 ± 2.98	*68.67 ± 2.17*	**61.28 ± 0.83**
	12	204.62 ± 66.00	148.64 ± 6.48	179.77 ± 4.59	193.90 ± 27.37	395.04 ± 47.19	117.74 ± 2.20	138.38 ± 10.72	116.31 ± 3.40	117.57 ± 10.47	117.75 ± 5.94	156.94 ± 10.10	130.99 ± 10.97	256.99 ± 102.95	117.74 ± 2.20	107.61 ± 17.22	351.56 ± 43.28	129.32 ± 10.78	209.83 ± 8.49	83.24 ± 3.42	*79.72 ± 2.02*	**69.72 ± 1.07**
	Avg	193.46 ± 66.47	129.32 ± 4.73	168.65 ± 5.79	150.20 ± 16.33	388.14 ± 34.47	107.35 ± 2.91	129.96 ± 13.13	101.75 ± 4.62	105.03 ± 9.62	106.12 ± 8.44	143.57 ± 7.65	121.28 ± 11.38	240.99 ± 101.10	107.35 ± 2.91	107.21 ± 22.05	332.73 ± 48.21	113.86 ± 10.61	182.86 ± 11.55	74.34 ± 3.20	*68.08 ± 2.30*	**61.84 ± 0.87**